What Risk Disclosures Reveal: Mining SEC 10-K Filings with Retrieval-Augmented Generation

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SEC 10-K Filings · Apple & Google · 2023 – 2025

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RESEARCH QUESTION

What do 600 pages of risk disclosures reveal about how tech companies see themselves?

DATA

Six SEC 10-K annual filings: Apple & Google, fiscal years 2023–2025. Dense legal and financial prose; 54–70% of content chunks contain tables.

METHOD — RAG PIPELINE

- HTML extraction from SEC EDGAR preserving PART → Item → Note hierarchy
- Sentence-boundary chunking, 800-token target → 578 chunks
- Local embeddings: nomic-embed-text-v1.5 (768 dims)
- Hybrid retrieval: 50% dense + 40% BM25 + 10% keyword overlap
- Answer generation with a local LLM (qwen2.5:3b)

WHY HTML, NOT PDF

Early PDF extraction failed on broken font encodings and borderless tables, forcing OCR and vision-AI fallbacks. SEC's HTML filings eliminated all of it — structure came for free.

ANALYSIS

Keyword frequency analysis (TF) on Item 1A — Risk Factors — per company, per year. Compared vocabularies across companies and tracked term trends 2023 → 2025.

LIMITATIONS

Two companies only; frequency-based keywords (no topic modelling); no chunk overlap; small local LLM context window. Trends are indicative, not statistically tested.

QUERIES ANSWERED

Q: Apple net sales 2023?
A: $383,285 million
Q: Services growth '23-'25?
A: $85,200M → $109,158M (+28%)
Q: Bitcoin holdings 2025?
A: Not found - not disclosed

MAIN FINDING

Risk language mirrors business models.

Apple talks supply chains.

Google talks regulation.

Two tech companies, different fears.

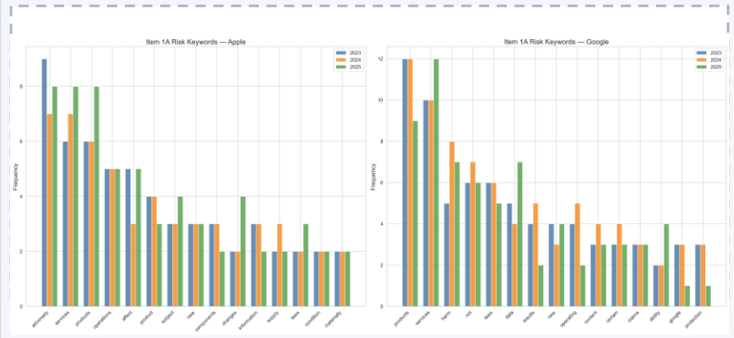
In three years of risk disclosures, Apple's most frequent terms are components, supply, products, operations — the vocabulary of a company whose fate hangs on factories and logistics. Google's are data, laws, content, protection — the vocabulary of a platform under regulatory siege.

The AI signal: Google's "results" keyword climbs steadily through 2024–2025 — as generative AI reshapes how search results are produced and monetised, the uncertainty shows up in its risk language first.

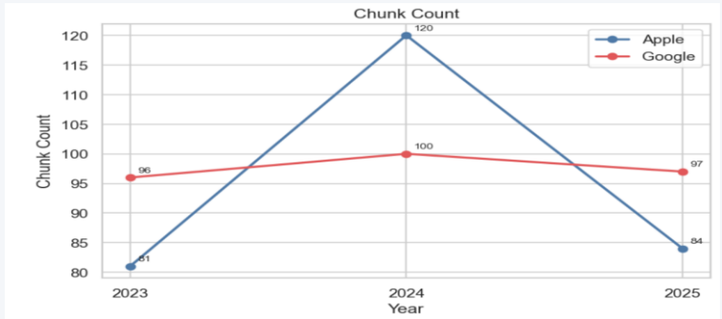
Found by asking, not reading: a fully local RAG pipeline over 578 structured chunks surfaces these patterns with source-cited answers — no cloud APIs, no data leaving the machine.

Scan for the full pipeline,
EDA notebook & documentation

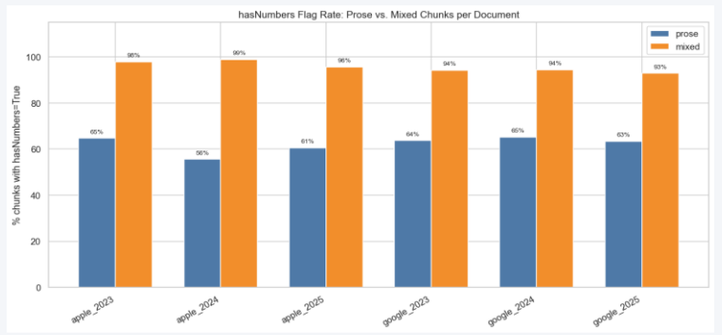
EVIDENCE & FIGURES



Apple: components, supply, products. Google: data, laws, protection. Two different worlds in the same document type.



Apple's 2024 filing spikes in size and table density, then drops back. Google stays flat — making Apple the harder corpus for a generalised pipeline.



Apple: 81 → 120 → 84 chunks. Google: 96 → 100 → 97. Stability is a corpus property, not a given.